Multidimensional Scaling and Distance Embedding

Reading: Izenman, Chapter 13

The main reference for these slides is Dr. Markus Kalisch's lecture notes:

Multidimensional Scaling lecture notes

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Agenda

Main Idea

Classical Multidimensional Scaling

Non-metric Multidimensional Scaling

Learning Objectives

By the end of this lecture, you should be able to:

- **Explain the goal of multidimensional scaling (MDS)** and distinguish it from PCA.
- **Describe classical MDS** and explain how pairwise distances are converted into a low-dimensional Euclidean representation.
- **Interpret eigenvalues and goodness of fit** when selecting the dimension of a classical MDS representation.
- **Explain non-metric MDS** and interpret the role of disparities, rankings, and STRESS.

PCA vs. Multidimensional Scaling (MDS)

- **Principal Component Analysis (PCA)**
  - **Input:** Data matrix of n observations and p variables.
  - **Goal:** Find a low-dimensional linear projection that maximizes retained variance.
- **Multidimensional Scaling (MDS)**
  - **Input:** Pairwise distances or dissimilarities $\boldsymbol{D} = (d_{ij})$.
  - **Goal:** Represent observations in $\mathbb{R}^r$ while preserving their dissimilarity relationships.

Key distinction: PCA starts with variables; MDS starts with relationships between observations.

Why Does MDS Matter?

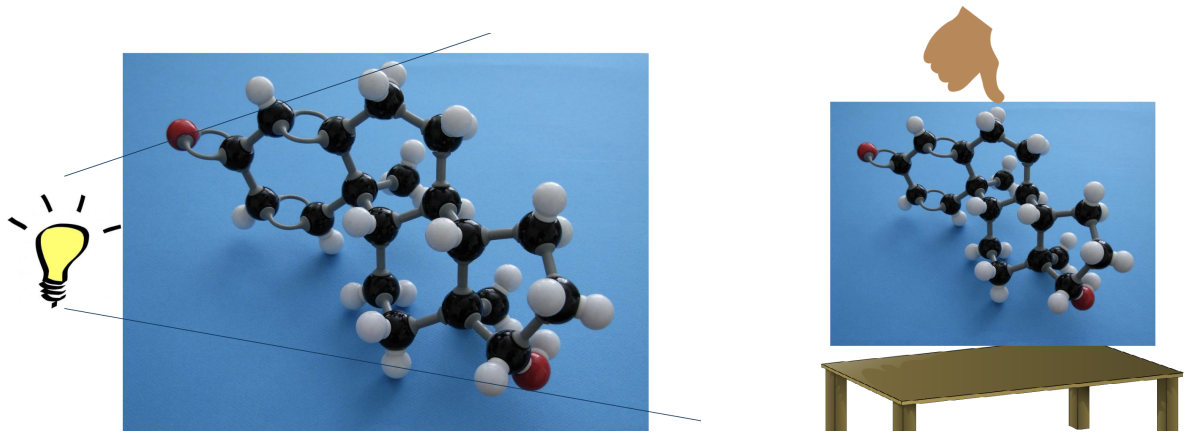
- Sometimes we know how **similar or different** observations are, but not their original features.
- MDS converts pairwise dissimilarities into a **low-dimensional representation**.
- The resulting map helps reveal **similarities, groups, and relative positions**.

Applications:

- **Marketing:** Visualizing customer or product preferences.
- **Geography:** Reconstructing relative locations from distances.
- **Data exploration:** Visualizing similarity-based relationships.

Basic Idea of Multidimensional Scaling

Goal: Find a low-dimensional configuration whose inter-point relationships reflect the observed dissimilarities.



Source: Dr. Markus Kalisch's Lecture Notes on MDS

Two common approaches:

- **Classical MDS:** Uses eigendecomposition of a matrix derived from squared dissimilarities.
- **Non-metric MDS:** Preserves the ranking of dissimilarities rather than their exact values.

Classical MDS: Can We Reconstruct a Map?

Example: Pairwise distances among 10 major U.S. cities.

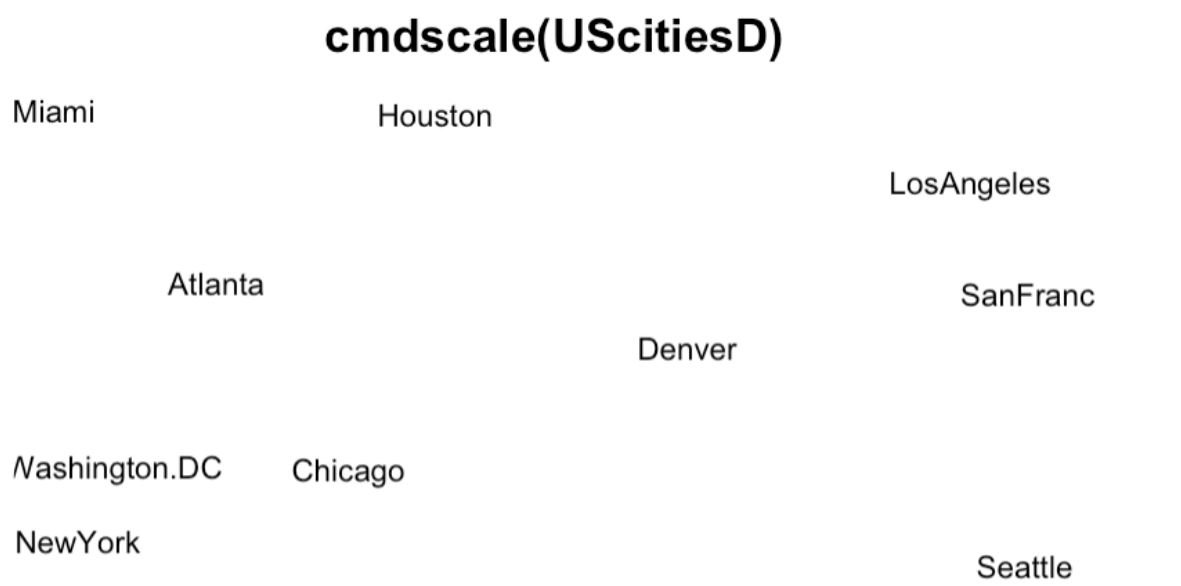
```
> UScitiesD
              Atlanta Chicago Denver Houston LosAngeles Miami NewYork SanFrancisco Seattle
Chicago           587
Denver           1212     920
Houston           701     940    879
LosAngeles       1936    1745    831    1374
Miami             604    1188   1726     968       2339
NewYork           748     713   1631    1420       2451  1092
SanFrancisco     2139    1858    949    1645        347  2594    2571
Seattle          2182    1737   1021    1891        959  2734    2408          678
Washington.DC     543     597   1494    1220       2300   923     205         2442    2329
```

- **Given:** Distances between cities, but not their geographic coordinates.
- **Question:** Can we recover their relative locations using only these distances?
- **Idea:** Classical MDS finds a low-dimensional configuration that approximately preserves pairwise distances.

Classical MDS: Reconstructing the U.S. Map

Using only pairwise distances, classical MDS produces the following two-dimensional configuration.

```
loc <- cmdscale(UScitiesD)
x <- loc[, 1]; y <- loc[, 2]
plot(x, y, type = "n", xlab = "", ylab = "", asp = 1,
     axes = FALSE, main = "cmdscale(UScitiesD)")
text(x, y, rownames(loc), cex = 0.8)
```

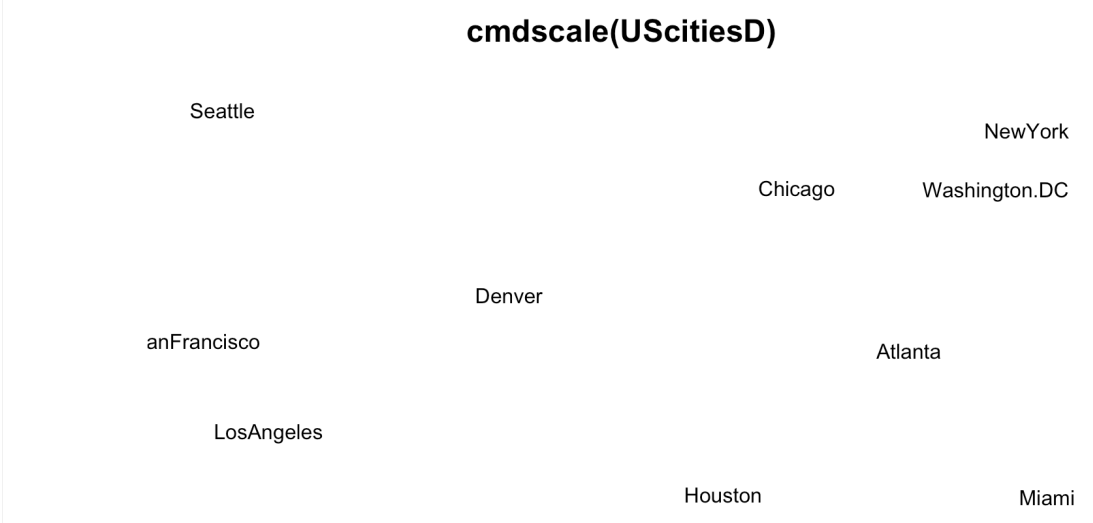


Question: Why does the map look different from the familiar U.S. map?

Classical MDS: Orientation Is Arbitrary

A reflection of the MDS coordinates produces a more familiar geographic orientation.

```
# Flip Axes
x1 <- -loc[, 1]; y1 <- -loc[, 2]
plot(x1, y1, type = "n", xlab = "", ylab = "", asp = 1,
     axes = FALSE, main = "cmdscale(UScitiesD)")
text(x1, y1, rownames(loc), cex = 0.8)
```



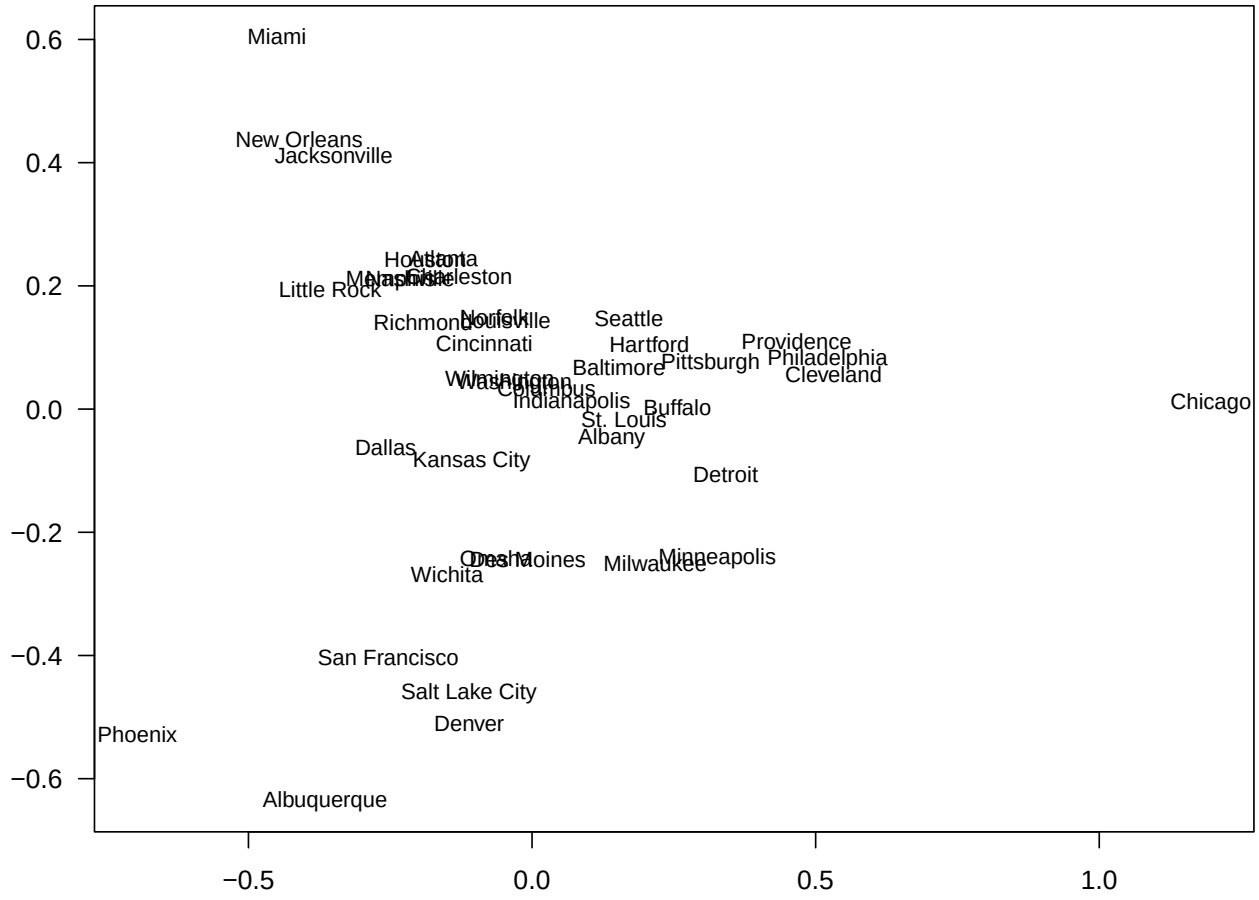
- Translation, rotation, and reflection **preserve all pairwise distances**.
- Therefore, MDS coordinates are **not unique in orientation or location**.

Air Pollution in U.S. Cities: Why Standardize?

```
> summary(dat)
      SO2              temp            manu            popul
 Min.   :  8.00   Min.   :43.50   Min.   :  35.0   Min.   :  71.0
 1st Qu.: 13.00   1st Qu.:50.60   1st Qu.: 181.0   1st Qu.: 299.0
 Median : 26.00   Median :54.60   Median : 347.0   Median : 515.0
 Mean   : 30.05   Mean   :55.76   Mean   : 463.1   Mean   : 608.6
 3rd Qu.: 35.00   3rd Qu.:59.30   3rd Qu.: 462.0   3rd Qu.: 717.0
 Max.   :110.00   Max.   :75.50   Max.   :3344.0   Max.   :3369.0
      wind            precip          predays
 Min.   : 6.000   Min.   : 7.05   Min.   : 36.0
 1st Qu.: 8.700   1st Qu.:30.96   1st Qu.:103.0
 Median : 9.300   Median :38.74   Median :115.0
 Mean   : 9.444   Mean   :36.77   Mean   :113.9
 3rd Qu.:10.600   3rd Qu.:43.11   3rd Qu.:128.0
 Max.   :12.700   Max.   :59.80   Max.   :166.0
```

- Variables such as `manu` and `popul` have much larger numerical scales than `wind`.
- Without standardization, variables with larger variability may **dominate Euclidean distances**.
- **Solution:** Standardize variables before computing pairwise distances.

Air Pollution: MDS After Standardization



- Distances now reflect differences across **all standardized variables**.
- Cities closer together have more similar multivariate profiles.
- **Key takeaway:** Scaling changes distances and therefore the MDS configuration.

Classical MDS: From Distances to Inner Products

- **Input:** Pairwise Euclidean distances $\boldsymbol{D} = (d_{ij})$ among n observations.
- **Key idea:** Convert squared distances into inner products of **centered coordinates**.

$$\boxed{\boldsymbol{B} = -\frac{1}{2}\boldsymbol{H}\boldsymbol{D}^{(2)}\boldsymbol{H}}$$

$$\boldsymbol{H} = \boldsymbol{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T, \qquad \boldsymbol{D}^{(2)} = (d_{ij}^2).$$

- $\boldsymbol{H}$ centers the observations; $\boldsymbol{B}$ is their **Gram matrix**.
- Equivalently,

$$b_{ij} = -\frac{1}{2}\left(d_{ij}^2 - \overline{d^2}_{i\cdot} - \overline{d^2}_{\cdot j} + \overline{d^2}_{\cdot\cdot}\right).$$

Classical MDS: Recovering Coordinates

- For Euclidean distances, the centered Gram matrix $\boldsymbol{B}$ is **positive semidefinite**.

- Compute its eigendecomposition:

$$\boldsymbol{B} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T, \qquad \lambda_1 \geq \cdots \geq \lambda_q > 0,$$

  where $q = \operatorname{rank}(\boldsymbol{B})$.

- **Recover the coordinates:**

$$\boxed{\boldsymbol{X} = \boldsymbol{V}_q \boldsymbol{\Lambda}_q^{1/2}} \qquad \boldsymbol{B} = \boldsymbol{X}\boldsymbol{X}^T.$$

- The recovered configuration is unique **up to rotation, reflection, and translation**.

Connection to PCA: Classical MDS and PCA give equivalent coordinates when MDS uses Euclidean distances.

Classical MDS: Low-Dimensional Representation

- **Dimension reduction:** Retain the r largest positive eigenvalues and their eigenvectors ($r < q$).

$$\boxed{\boldsymbol{X}_r = \boldsymbol{V}_r \boldsymbol{\Lambda}_r^{1/2}}$$

- **Goodness of fit:** Proportion of total positive eigenvalue magnitude retained:

$$\mathrm{GOF}(r) = \frac{\sum_{i=1}^{r} \lambda_i}{\sum_{i=1}^{q} \lambda_i}.$$

- **Optimality:** $\boldsymbol{X}_r \boldsymbol{X}_r^T$ is the best rank-r approximation to $\boldsymbol{B}$ in Frobenius norm.

Takeaway: Larger eigenvalues represent more of the configuration's squared-distance structure.

Classical MDS: Pros and Cons

- + **Exact recovery** when the distances are Euclidean and the embedding dimension is sufficiently large.
- + **Closed-form solution** via eigendecomposition, without iterative optimization.
- + **Computationally efficient** for moderate datasets.
- – **Non-Euclidean dissimilarities** may produce negative eigenvalues, preventing exact Euclidean representation.
- – **Limited scalability**: eigendecomposition of an $n \times n$ matrix becomes costly for large n.

Non-metric MDS: Why Rankings Matter

Example: Comparing similarities among animal species.

- We may judge that a **lion and tiger** are more similar than a **lion and elephant**.
- Such comparisons provide a meaningful **ordering of dissimilarities**, even without precise numerical distances.
- **Non-metric MDS** constructs a low-dimensional map that preserves these rankings as closely as possible.

Key idea: Relative similarities matter more than exact distances.

Non-metric MDS: How It Works

- Let δ_{ij} be the observed dissimilarity and d_{ij} the distance in the MDS map.
- Find a **monotone transformation** $\theta(\cdot)$ that preserves dissimilarity rankings:

$$\delta_{ij} < \delta_{kl} \quad \Longrightarrow \quad \theta(\delta_{ij}) \leq \theta(\delta_{kl}).$$

- Choose the configuration and transformation to minimize **STRESS**:

$$\boxed{S = \frac{\sum_{i<j}[\theta(\delta_{ij}) - d_{ij}]^2}{\sum_{i<j} d_{ij}^2}}$$

- The transformed values $\hat{d}_{ij} = \theta(\delta_{ij})$ are called **disparities**.

Goal: Find distances that closely match the disparities.

Non-metric MDS: Pros and Cons

- + **Suitable for ordinal data** and subjective dissimilarity judgments.
- + **Preserves rankings** rather than requiring meaningful distance magnitudes.
- + **Invariant to strictly increasing transformations** of dissimilarities.
- – **Iterative optimization** can be computationally demanding and sensitive to initialization.
- – **Distance magnitudes are not preserved**; solutions may converge to local optima.

Congressional Voting: Measuring Dissimilarity

Data: Voting records of 15 New Jersey members of Congress on 19 environmental bills (Romesburg, 1984).

```
> voting[1:6, 1:6]
```

	Hunt(R)	Sandman(R)	Howard(D)	Thompson(D)	Freylinghuysen(R)	Forsythe(R)
Hunt(R)	0	8	15	15	10	9
Sandman(R)	8	0	17	12	13	13
Howard(D)	15	17	0	9	16	12
Thompson(D)	15	12	9	0	14	12
Freylinghuysen(R)	10	13	16	14	0	8
Forsythe(R)	9	13	12	12	8	0

- Each entry counts how many times two members **voted differently**.
- **Smaller values** indicate more similar voting records.

Question: Do members of the same party tend to vote similarly?

Congressional Voting: Non-metric MDS in R

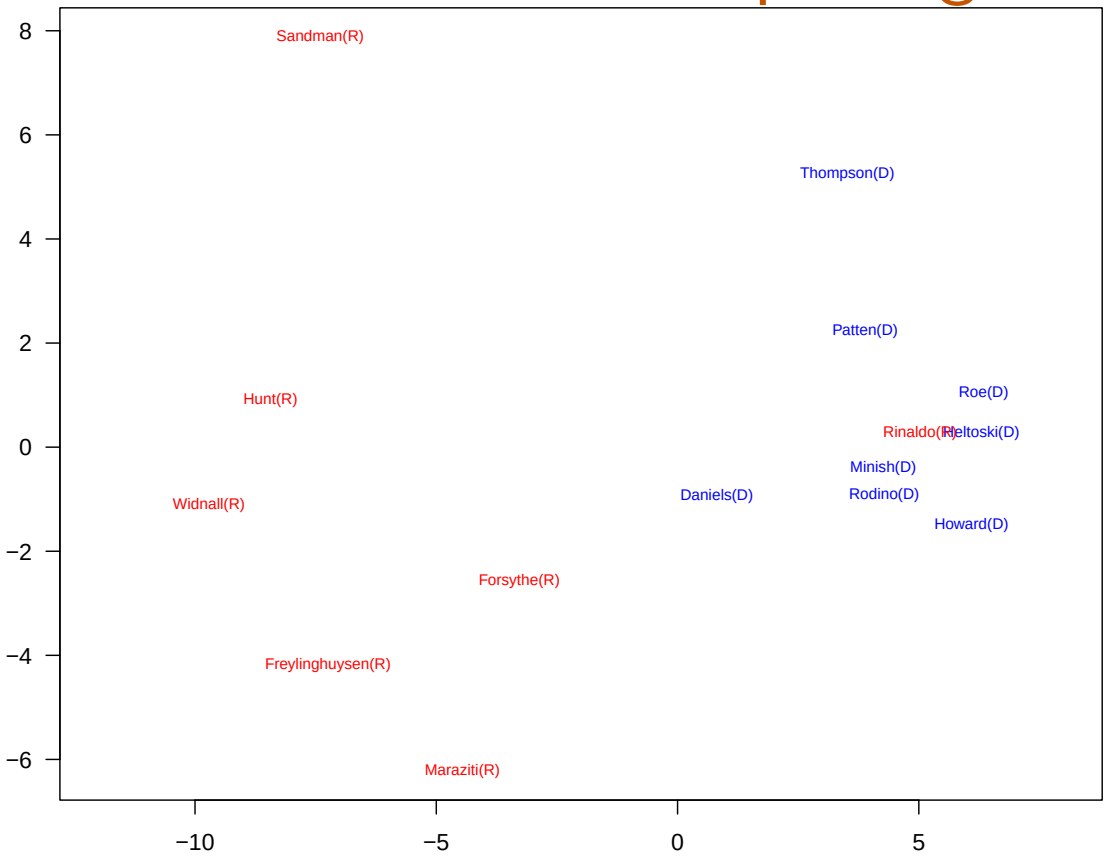
Method: Use `isoMDS()` to find a two-dimensional configuration.

- `d`: Pairwise voting dissimilarities.
- `k = 2`: Two-dimensional embedding.
- `cmdscale(d, k = 2)`: Classical MDS initialization.

```
library(MASS)
voting_mds <- isoMDS(voting, k = 2)
str(voting_mds)
par(las = 1, mar = c(2, 2, 0.5, 0.5))
plot(voting_mds$points, type = "n", xlim = c(-12, 8),
     xlab = "", ylab = "")
text(voting_mds$points, labels = rownames(voting_mds$points),
     cex = 0.7, col = col)
```

Goal: Find a map that preserves the ranking of voting dissimilarities.

Non-metric MDS: Interpreting the Voting Map



- **Proximity:** Nearby members tend to have similar voting records.
- **Separation:** Distinct groups may indicate different voting patterns.
- **STRESS:** Measures how well the two-dimensional map represents the dissimilarity rankings.

Classical vs. Non-metric MDS

	Classical MDS	**Non-metric MDS**
Input	Distance matrix	Dissimilarity matrix
Goal	Preserve Euclidean distance structure	Preserve dissimilarity rankings
Method	Double-centering and eigen-decomposition	Iterative STRESS minimization
R function	`cmdscale()`	`MASS::isoMDS()`

Key distinction: Distance magnitudes vs. dissimilarity rankings.

Summary: Choosing a Dimensionality Reduction Method

Which method would you choose?

- **PCA:** We have numerical features and want to summarize their variation using a few linear components.
- **Classical MDS:** We have meaningful pairwise distances and want a low-dimensional map of the observations.
- **Non-metric MDS:** We have ordinal dissimilarities and care more about their rankings than exact magnitudes.

Looking ahead: What if the data lie on a curved, nonlinear structure?

Next week: Nonlinear PCA and Manifold Learning